

Skills Summary

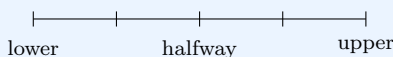
Rounding Multi-Digit Numbers to Named Places

Use this reference while completing your worksheet or when studying.

Rounding is a choice between two numbers: the multiple of the named place just below, and the one just above. The halfway number decides which one wins. Numbers here are written without commas.

1. Rounding on a number line

Three steps: find the benchmark below and the benchmark above, mark the halfway number (lower benchmark + half the place), then see which side the number falls on. On halfway exactly, round up.



Example: Round 6382 to the nearest thousand.

Step 1. The named place is thousands, so the only two candidates are the thousands on either side — 6000 and 7000.

Step 2. Halfway is 6500, and 6382 has not reached it, so the lower benchmark wins.

6000

Try it: Round 2471 to the nearest hundred.

2500 :check

2. Rounding to any named place

Digit rule (the number line, written short): look at the digit *one place to the right* of the named place. If it is 5 or more, round up; if it is 4 or less, round down. Every digit right of the named place becomes 0; digits to the left stay put unless a round-up carries into them.

Example: Round 35728 to the nearest thousand.

Step 1. The thousands digit is 5, so look one place to its right — the hundreds digit — because that is where the halfway test lives.

Step 2. The hundreds digit is 7, which is 5 or more, so round the thousands up from 35 to 36 and make the last three digits zeros.

36000

Try it: Round 9963 to the nearest hundred. (Watch the carry.)

10000 :check

3. Estimating a sum or difference

Rule: round *each* number to the named place first, then add or subtract the rounded numbers. Never round the answer you already computed — an estimate is meant to be quick and mental.

Example: Estimate $587 + 319$ to the nearest hundred.

Step 1. The point is a mental sum, so round both numbers before adding anything.

Step 2. 587 is past 550, so it becomes 600; 319 is before 350, so it becomes 300; then $600 + 300$.

900

Try it: Estimate $4128 + 3675$ to the nearest thousand.

check: 7803

4. Rounding numbers from real situations

Rule: read which place the question names before you round anything — “to the nearest thousand” and “to the nearest ten thousand” give different answers for the same count. Keep the unit (people, metres, dollars) in the answer, and remember a rounded count is a report, not the exact number.

Example: A city library lent 18642 books last year. Report that to the nearest thousand.

Step 1. The question names thousands, so the candidates are 18000 and 19000 — not ten thousands.

Step 2. Halfway is 18500 and 18642 is past it, so round up.

19000 books

Try it: A stadium seats 7409 people. Report that to the nearest hundred.

check: 7400